



USER GUIDE

Controller Router

Assign physical controllers and VirtualGlove to the right game players.

2026 EDITION / IAIN BENNETT / MIT LICENSE

Controller Router lets you choose which gamepad controls each player, and which systems use those choices. It also brings VirtualGlove and Buddy together in one place.

Already happy with your controls? Keep your saved choices. You only need this guide when you want to change a player, choose a system, or troubleshoot a pad.

Open Setup

1. In your browser, open the controller address shown by its installer.
2. Choose **Players and Systems** on the Controller Router page.
3. Select your paired console under **Console connection**.

You will see **Players** first, then **Systems**. Each console keeps its own settings. If no console appears, choose **Pair console** in Controller Router Setup first.

Choose who plays

Before assigning a gamepad, configure its buttons in EmulationStation, your console's game menu.

1. Exit any running game.
2. In Router **Setup > Players**, find your gamepad by name.
3. Choose **Player 1**, **Player 2**, **Player 3**, or **Player 4**. Choose **Unassigned** if you do not want that pad to join a Router player.
4. Choose your gesture player under **VirtualGlove player**, or leave it **Unassigned**.
5. Choose **Save assignments**.
6. Launch a game to try the new choices.

Keep a physical gamepad assigned to **Player 1** for the usual menu and exit controls. Several pads can share one player; a pad can belong to only one player. Buddy stays on **Player 2** for Gyromite's red and blue gates.

A game must support additional players for Player 3 or Player 4 to do anything. Assigning four pads does not turn a two-player game into a four-player game.

Choose which systems use Router

Want Router for NES but your normal controls for Mega Drive / Genesis? You can choose each system separately.

1. With the game closed, open **Setup > Systems**.
2. Under **Routing selection**, choose **Choose systems individually**.
3. For each listed system, choose one of the following options.
4. Choose **Save assignments**, then launch your next game.

Choose	What happens in a game
Controller Router	The system uses the player assignments you saved in Players , with the buttons you configured in EmulationStation.

Choose	What happens in a game
My existing setup	The system uses its normal controller setup. Router adds no player-routing changes for that game.

Buddy's games and VirtualGlove require Controller Router enabled for NES. You can still choose **My existing setup** for NES when you want to use your own controls instead.

New Router setups start with **NES only**. Upgrades keep your choices. **All Libretro systems, including newly added systems** applies Router to every supported RetroArch system. In individual mode, a newly discovered system starts with **My existing setup**.

The list covers systems that offer supported RetroArch emulators. Separate emulators, such as a standalone Amiga emulator, keep their own controls. Your choice also stays saved if a system is temporarily unavailable.

Check a gamepad

1. In Router Setup, choose **Test inputs**.
2. Press a direction or button on the pad during the next five seconds.
3. Read the result, then repeat for another pad if needed.

No button presses detected can simply mean you did not press anything during the test. Try again before changing settings.

Status	What to do
Connected	The pad is available. Try Test inputs to identify it.
Unavailable	Turn on or reconnect the saved pad. If you replaced it, configure and assign the new one.
Mapping refreshed or updated	Router found your newer EmulationStation button setup. Exit and relaunch the game to use it.

Reload brings back the saved choices and discards unsaved edits. **Restore previous** restores the preceding saved setup. Exit the game before saving or restoring.

Start a game or choose an app

Launch a registered game on your paired console. Router selects VirtualGlove or R.O.B. Vision automatically, so you do not need a browser open for the controller to join the game. Only that app supplies game input and display cues. When the game ends, the Matrix returns to its neutral animation.

To open an app yourself, visit the controller address. With one app installed, it opens directly. With both installed, choose VirtualGlove or R.O.B. Vision. **Apps** in either app returns to the chooser; it appears only when both are installed. Finish a game before changing apps manually. Both apps stay available in the background.

After a reboot, the controller waits for a game or your choice. You do not need to reinstall or pair again.

Get back to your game

My pad works in the game menu but not in the game. Try another connected pad: yours may be assigned to Player 2. Exit the game, check **Players**, and use **Test inputs** to identify the pad you want on Player 1. Save and relaunch.

A wireless pad went to sleep. Wake it or reconnect it. Router keeps its player controllers connected and returns the pad to its saved player. If the Router service itself restarted, exit and relaunch the game once it is ready.

My buttons changed after I remapped them. Exit the game, confirm the new buttons in EmulationStation, and relaunch. Your player assignment stays saved.

I cannot save. Finish the running game first. If another page changed the settings, choose **Reload**, make your choices again, and save.

The wrong app is showing. Check that the game is registered in the right app and that the console is paired with this controller. Exit the game before making changes.

Play with VirtualGlove

After the game starts, rest your hand at its saved centre once, then make your first gesture. A physical Player 1 pad supplies menu and exit hotkeys; glove Select remains the game's Select button.

For joystick versus native Super Glove Ball controls, see [Input Modes](#). For camera, pairing, or game problems, see [Troubleshooting](#). Advanced routing details are in the [Configuration Reference](#).

Pair once for both apps

Pair your console once through Controller Router. VirtualGlove and R.O.B. Vision receive their own private credentials automatically when installed on both devices. Installing the other app later adds its access without another pairing. No SSH username or password is required.

Finish the game before pairing, changing app access, or removing a connection. Each console connects to one Controller Router installation at a time. Connecting it to another requires a new console code and Matrix confirmation.

1. Open **Apps > Setup > Pair console**. Both product Setup pages have an **Open Pair console** link to this same page.
2. Open the secure address printed by the controller installer, using its `.local` name or LAN IP. Pairing uses HTTPS port **8444**.
3. Before accepting the local certificate, compare the browser's SHA-256 fingerprint with the fingerprint printed by the controller installer. During confirmation, its beginning also appears after **ID** on the Matrix. Stop if they differ.
4. Enter the console hostname or IP address and paste its complete **CR1 connection code**. The console installer prints this single-use code; it lasts five minutes.
5. Choose **Continue**, read the six Matrix digits after **PN**, and enter them within two minutes.
6. Choose **Connect**. Wait for **Connected** and check each app's readiness below it.

Check or repair a connection

Open **Pair console > Your consoles**. **Connected** means Router has verified the console connection. **Unavailable** means it could not reach the console. **Needs attention** means the certificate, identity, or app setup needs review. App readiness is shown separately.

Choose **Check and repair connections** after reconnecting a device or installing another app. Use **Disable** beside an app to remove only its access, or **Remove console** to remove the whole connection. Finish any game first. A certificate change requires a fresh pairing; do not ignore the mismatch.

For another code, rerun the console installer or its pairing command. Existing game filenames and player assignments remain saved. Incorrect, expired, or already-used codes require a new window; five incorrect Matrix confirmations lock the current window.